

D'Decor Workboard

Internal Project Management Platform

A fully self-hosted, production-grade Monday.com replica built for D'Decor's internal teams. Powered by React, Node.js, and PostgreSQL, deployed on a private Ubuntu server with Docker Compose, Azure AD SSO, corporate SSL, and automated workflows — fully owned by D'Decor.

Active Users

79

Live Boards

57

Docker Containers

5

Security Score

7.5/10

<https://nocode.ddecor.com>
Production URL

Table of Contents

1		
.	Platform Overview	3
2		
.	Technical Architecture	4
3		
.	Infrastructure & Deployment	5
4		
.	Security	6
5		
.	Feature Set	7
6		
.	User & Board Statistics	9
7		
.	Integrations	10
8		
.	Developer & Operational Guide	11
9		
.	Roadmap & Pending Items	12
10		
.	Appendix — Hard-Won Lessons	13

1. PLATFORM OVERVIEW

What is D'Decor Workboard?

D'Decor Workboard is a fully self-hosted, production-grade project management platform built exclusively for D'Decor Home Fabrics Pvt. Ltd. It replicates the core functionality of Monday.com with a familiar, polished interface — while being 100% owned, hosted, and controlled by D'Decor's IT department. No vendor lock-in, no per-seat SaaS billing, no external data exposure.

The platform was built iteratively from the ground up using modern open-source technologies and is now live in production at <https://nocode.ddecor.com>, serving 79 users across 57 boards.

Key Objectives

Self-Hosted	All data stays within D'Decor's own infrastructure. No data is shared with third-party SaaS platforms.
Monday.com Parity	Mirrors Monday.com's UX and features — boards, items, subitems, status columns, assignments, and more.
Cost Efficiency	Eliminates per-seat SaaS licensing costs that would scale with headcount.
Customizability	Fully open to D'Decor-specific workflows, integrations (Teams, Azure AD), and internal reporting needs.
Security	Corporate SSO, API keys, role-based access, and HTTPS with automated SSL renewal.

Platform Identity

Platform Name	D'Decor Workboard
Production URL	https://nocode.ddecor.com
Server Hostname	honocode (Ubuntu Server)
Server Path	/data/workday/
Git Repository	github.com/anuptechin/workday (Private)
Dev Environment	C:\Dev\DdecorMonday (Windows)
Status	LIVE — Production
Launch Date	2024 (Iterative development)
Managed By	D'Decor IT Department

2. TECHNICAL ARCHITECTURE

Technology Stack

Layer	Technology	Details
Frontend	React + Vite	SPA, Figtree font, Light/Dark/System themes, mobile-responsive
Backend	Node.js + Express	REST API, JWT auth, API key middleware, session management
Database	PostgreSQL	Relational DB, structured board/item/user data, automated backups
Containerisation	Docker Compose	5 containers: postgres, node, react, nginx, certbot
Web Server	Nginx	Reverse proxy, SSL termination, shared proxy at /data/nginx-proxy/
SSL	Let's Encrypt + Certbot	Automated certificate issuance, twice-daily renewal checks
Authentication	Azure AD (Microsoft 365)	SSO via OAuth2/OIDC, integrated with corporate tenant
Email	Microsoft 365 SMTP	monday@ddecor.com, SMTP AUTH enabled via PowerShell
Version Control	Git (GitHub)	main branch (app code) + production branch (server configs)

Container Architecture

The application runs as five Docker containers orchestrated by Docker Compose, all started and managed from **/data/workday/**. A shared Nginx reverse proxy at **/data/nginx-proxy/** handles routing for all apps hosted on the same server.

Container	Image / Role	Key Notes
postgres	PostgreSQL — Primary database	Persistent volume, automated twice-daily backups
node	Node.js — Express API server	Handles auth, board logic, email triggers
react	React/Vite — Frontend SPA	Built with Vite, served via Nginx
nginx	Nginx — Internal app router	Serves frontend, proxies /api to node container
certbot	Let's Encrypt — SSL certs	Auto-renewal, certificate stored in shared volume

Data Flow

Browser → Nginx Proxy (nocode.ddecor.com) → Nginx Container → React SPA (static) + Node.js API (/api/*) → PostgreSQL. Azure AD handles authentication before any board data is served. Email notifications are dispatched by the Node.js service via Microsoft 365 SMTP.

3. INFRASTRUCTURE & DEPLOYMENT

Server Details

Hostname	honocode
OS	Ubuntu Server (LTS)
Current RAM	4 GB (8 GB upgrade recommended)
Application Root	/data/workday/
Other Apps on Server	/data/monday/, /data/pos-ddecor/, /data/nginx-proxy/
Shared Nginx Proxy	/data/nginx-proxy/ (routes all apps on server)
SSL Certificates	Let's Encrypt via Certbot container
Backup Schedule	Twice-daily automated PostgreSQL dumps

Deployment Process

A one-command deploy script at `/data/workday/deploy.sh` handles the entire deployment lifecycle. All Docker commands must be run from `/data/workday/` — never from the home directory.

Standard deployment sequence:

- **Step 1:** Pull latest code from GitHub (main branch)
- **Step 2:** Build new Docker images
- **Step 3:** docker compose down (stops all containers)
- **Step 4:** docker compose up -d (starts with new images + env vars)
- **Step 5:** Hard-refresh browser with Ctrl+Shift+R to clear cached assets

Critical Deployment Rules

	docker compose restart does NOT apply new volume mounts or environment variables — always use down && up -d
	Windows-to-Docker builds lose execute permissions on node_modules/.bin/ files — fixed via chmod +x node_modules/.bin/vite in Dockerfile
	.env special characters (@ ! #) must be wrapped in single quotes
	VPN on laptop can make the production site appear down — always check VPN state before diagnosing
	gc.auto=0 is set in dev environment (C:\Dev\DdecorMonday) to prevent Git garbage collection loops

Git Branch Strategy

Branch	Purpose	Notes
main	Application source code	Active development, all feature branches merge here
production	Server-specific configs	Diverged from main — monitor for .gitignore conflicts

4. SECURITY

Security Overview

The platform has undergone a formal security review with a current score of **7.5/10**. Key security layers include Azure AD SSO, API key authentication, HTTPS enforcement, and role-based access controls. Two items (M-5 and L-2) remain pending as quick wins.

Implemented Security Controls

Control	Status	Details
Azure AD SSO	✓ Live	Microsoft 365 OAuth2/OIDC, corporate tenant — no local password auth for SSO users
API Key Authentication	✓ Live	API keys implemented for programmatic access, key middleware on protected routes
HTTPS / SSL	✓ Live	Let's Encrypt cert via Certbot, auto-renewal, all HTTP redirected to HTTPS
JWT / Session Secrets	■ Pending	Placeholder values still in use — replacement with strong generated values pending
Role-Based Access	✓ Live	Board-level permissions, owner/member/viewer roles
PostgreSQL Backups	✓ Live	Twice-daily automated dumps, stored on server
Gmail IMAP Poller	■ Pending removal	Currently causing noisy logs — removal scheduled
SSL Renewal Monitoring	■ Pending	Manual currently — automated monitoring not yet set up

Azure AD SSO Configuration Notes

- Copy the Secret VALUE from the Azure portal — not the Secret ID (common mistake).
- SMTP AUTH must be enabled via PowerShell: `Set-CasMailbox -SmtpClientAuthenticationDisabled $false`
- Tenant ID, Client ID, and Client Secret are stored in the .env file.
- SSO callback URL must be whitelisted in the Azure AD app registration.

Pending Security Items

ID	Item	Priority	Effort
M-5	Replace JWT/Session placeholder secrets with strong generated values	MEDIUM	Quick
L-2	Remove Gmail IMAP poller — causing log noise and unnecessary attack surface	LOW	Quick
–	Set up SSL certificate renewal monitoring (Uptime Kuma recommended)	MEDIUM	Medium
–	Server RAM upgrade from 4GB to 8GB (stability under load)	MEDIUM	Infra

5. FEATURE SET

Core Features

Board Management

- Create, edit, delete, and archive boards
- Multiple board views (Table, Kanban — extensible)
- Saved filters per board — persist across sessions
- Board cloning — duplicate entire board structure
- Shareable board links — controlled external access
- Board-level permissions (owner, member, viewer roles)

Items & Subitems

- Create and manage items within boards
- Subitems — nested tasks under parent items
- Custom column types: Status, Text, Date, People, Number, Dropdown, Tags
- Item assignment to users (owner by name; owner by ID — next sprint)
- Inline editing across all column types
- Item comments and activity log

User Interface

- Monday.com-style UI with Figtree font
- Light / Dark / System theme support
- Collapsible sidebar navigation
- Mobile-responsive layout (phone and tablet)
- Real-time updates within sessions

Authentication & Users

- Azure AD SSO via Microsoft 365 (OAuth2/OIDC)
- 79 active users managed via Azure AD
- API Key authentication for programmatic access
- Role-based access control at board level

Email & Notifications

- Email automation via monday@ddecor.com (Microsoft 365 SMTP)
- Notifications triggered by item assignment, status change, due dates
- Email dispatched from Node.js backend via SMTP

Migration & Import

- Custom Monday.com migration tool at migration-tool/
- Uses Monday.com GraphQL API to pull board/item data
- Imports boards, items, columns, and user assignments

PWA & Mobile

Phase 1 PWA is complete (installable as web app on mobile). Phase 2 is pending and will include native push notifications and enhanced offline support.

Planned Features (Next Sprint / Roadmap)

Feature	ID	Sprint
Owner by ID — resolve user assignments via ID not just name	M-1	Next Sprint
Microsoft Teams Bot integration	—	Roadmap
Director Daily Report UI — executive summary dashboard	—	Roadmap
PWA Phase 2 — push notifications	—	Roadmap
Uptime Kuma monitoring setup	—	Roadmap

6. USER & BOARD STATISTICS

Live Platform Metrics

As of 03 April 2026, the platform is live in production with the following metrics:



User Management

All users are managed through Azure Active Directory (Microsoft 365). User provisioning, deprovisioning, and group management happen at the Azure AD level, with the Workboard automatically reflecting the corporate directory. No local user database management is required.

Board Usage Notes

57 boards span multiple departments and project types. Boards can be cloned, shared externally via shareable links (with access controls), and migrated from Monday.com using the built-in migration tool. Board data is persisted in PostgreSQL and backed up twice daily.

Growth Indicators

Metric	Value	Notes
Total Users	79	Managed via Azure AD
Total Boards	57	Across all departments
Auth Method	Azure AD SSO	100% of users via SSO
API Key Users	Active	Programmatic integrations enabled
Migration Tool	Active	Monday.com → Workboard imports
Email Automation	Active	monday@ddecor.com live

7. INTEGRATIONS

Active Integrations

Azure Active Directory (SSO)

LIVE

Full OAuth2/OIDC integration with Microsoft 365 tenant. Users sign in with their corporate credentials. No separate passwords. Tenant ID, Client ID, and Client Secret configured in .env. Secret VALUE (not ID) must be copied from Azure portal.

Microsoft 365 SMTP (Email)

LIVE

Email sent from monday@ddecor.com via Microsoft 365 SMTP. SMTP AUTH enabled via PowerShell (Set-CasMailbox). Used for notifications, assignments, and status updates.

Monday.com GraphQL API (Migration)

LIVE

Custom migration tool uses Monday.com's GraphQL API to import boards, items, columns, and user data. Located at migration-tool/ in the repository.

Microsoft Teams Bot

PLANNED

Teams Bot integration is on the roadmap. Will allow Workboard notifications and actions directly within Microsoft Teams channels and personal chats.

Uptime Kuma (Monitoring)

PLANNED

Self-hosted uptime monitoring to be set up for production URL health checks, SSL expiry alerts, and container health monitoring.

8. DEVELOPER & OPERATIONAL GUIDE

Common Operational Commands

Start all containers	cd /data/workday && docker compose up -d
Stop all containers	cd /data/workday && docker compose down
Restart (with new env/volumes)	cd /data/workday && docker compose down && docker compose up -d
View logs (all)	docker compose logs -f
View logs (node only)	docker compose logs -f node
Check container status	docker compose ps
Run deploy script	cd /data/workday && bash deploy.sh
Access PostgreSQL	docker compose exec postgres psql -U -d
Manual DB backup	docker compose exec postgres pg_dump -U > backup.sql

Development Environment

Dev Path	C:\Dev\DdecorMonday (Windows)
Git Config	gc.auto=0 (prevents Git garbage collection loops)
Tooling	Claude Code (AI-assisted UI/feature development via prompts)
Build Tool	Vite (React fast builds, HMR in dev)
Post-deploy check	Always Ctrl+Shift+R hard-refresh in browser after deployment

Environment Variables (.env) Notes

- Special characters like @, !, # in passwords must be wrapped in single quotes in .env
- Azure AD requires: AZURE_TENANT_ID, AZURE_CLIENT_ID, AZURE_CLIENT_SECRET (the Value, not the ID)
- SMTP config: SMTP_HOST, SMTP_PORT, SMTP_USER (monday@ddecor.com), SMTP_PASS
- JWT_SECRET and SESSION_SECRET — currently placeholder values, pending replacement with strong generated strings (M-5)
- Database: DB_HOST=postgres (container name), DB_NAME, DB_USER, DB_PASS

9. ROADMAP & PENDING ITEMS

Sprint & Roadmap Overview

ID	Item	Category	Priority	Effort	Status
M-5	Replace JWT/Session placeholder secrets	Security	MEDIUM	Quick	Pending
L-2	Remove Gmail IMAP poller (noisy logs)	Security	LOW	Quick	Pending
M-1	Owner by ID (resolve users via ID)	Feature	HIGH	Sprint	Next Sprint
–	Microsoft Teams Bot integration	Integration	MEDIUM	Large	Roadmap
–	Director Daily Report UI	Feature	MEDIUM	Large	Roadmap
–	PWA Phase 2 — push notifications	Feature	LOW	Medium	Roadmap
–	Uptime Kuma monitoring setup	Infra	MEDIUM	Medium	Roadmap
–	SSL certificate renewal monitoring	Infra	MEDIUM	Quick	Roadmap
–	RAM upgrade 4GB → 8GB	Infra	MEDIUM	Infra	Recommended

Priority Legend

HIGH — Blocking or security critical	MEDIUM — Important, schedule soon	LOW — Nice to have
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10. APPENDIX — HARD-WON OPERATIONAL LESSONS

Lessons Learned in Production

These are real-world lessons discovered during development and production operation of the platform. They are documented here to prevent repeat issues.

Docker Must Run from /data/workday/

CRITICAL

All docker compose commands must be executed from /data/workday/. Running from ~ or another directory causes compose to look for the wrong docker-compose.yml and fails silently or with confusing errors.

Hard-Refresh After Deployment

IMPORTANT

After every deployment, always Ctrl+Shift+R hard-refresh in the browser. React's aggressive caching means old JS/CSS bundles will be served until force-refreshed, causing the app to appear broken or not updated.

'restart' vs 'down && up -d'

CRITICAL

docker compose restart does NOT apply new environment variables or volume mount changes. Always use docker compose down && docker compose up -d when any .env or compose config changes.

Windows Build — Execute Permissions Lost

IMPORTANT

When building Docker images on Windows and deploying to Linux, node_modules/.bin/ files lose their execute permissions. Fix: add chmod +x node_modules/.bin/vite (and other bin files) in the Dockerfile RUN step.

.env Special Characters

IMPORTANT

Special characters in .env values — particularly @, !, # in passwords and SMTP credentials — must be wrapped in single quotes. Unquoted, they break shell parsing and cause auth failures.

VPN Masking Production Issues

GOTCHA

The laptop VPN can make https://nocode.ddecor.com appear down when it is actually running fine. Always disable VPN or test from a different device before concluding the production server is down.

Azure AD — Secret Value vs Secret ID

CRITICAL

When configuring Azure AD SSO, the Azure portal shows both a Secret Value and a Secret ID. The AZURE_CLIENT_SECRET .env variable requires the Value, not the ID. Using the ID causes silent auth failures that are hard to debug.

Microsoft 365 SMTP Auth

IMPORTANT

Microsoft 365 SMTP requires explicitly enabling SMTP Authentication per mailbox via PowerShell: Set-CasMailbox -Identity monday@ddecor.com -SmtplibClientAuthenticationDisabled \$false. Without this, all SMTP sends fail with auth errors even with correct credentials.

production Branch Divergence

WATCH

The production Git branch has diverged from main, causing repeated .gitignore merge conflicts when syncing. Monitor this closely — consider re-basing or merging production back to main to reduce drift.

gc.auto=0 in Dev

IMPORTANT

Git garbage collection (gc.auto) is disabled in C:\Dev\DdecorMonday to prevent garbage collection loops that slow down Git operations in the Windows dev environment.

Document Information

Document Title	D'Decor Workboard — Platform Documentation
Version	Generated 03 April 2026
Classification	Confidential — Internal Use Only
Maintained By	D'Decor IT Department
Platform URL	https://nocode.ddecor.com
Repository	github.com/anuptechin/workday (Private)